



Exploring eDNA in SBDI

BioSyst.EU 2026



**SBDI
eDNA
team:**



**Tobias
Andermann
(UU)**



**Anders
Andersson
(KTH, PI)**



**Daniel
Lundin
(LnU)**



**Maria
Prager
(GU)**

What is SBDI ?



- National research infrastructure supported by the Swedish Research Council (VR)
- 11 universities and government agencies
- Host: Swedish Museum of Natural History
- **Swedish GBIF node:** national access point for **Open biodiversity data:**
 - Citizen science (bird watchers, amateur botanists...)
 - Systematic monitoring programs
 - Natural history collections
 - Research projects, including eDNA data
- **Tools and services** for interdisciplinary research on biodiversity and ecosystems



We mobilize biodiversity data



← → ↻ gbif.se ☆

Species occurrences published in GBIF by Swedish institutions

175,556,030

Occurrences

213

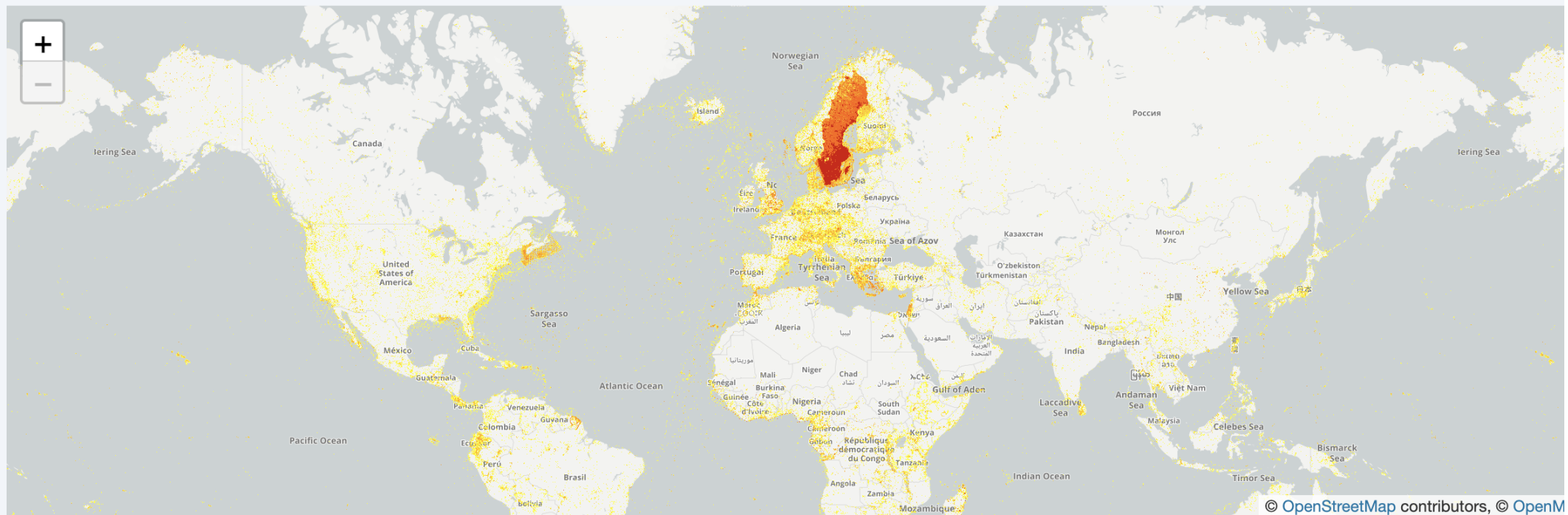
Datasets

251

Countries covered

43

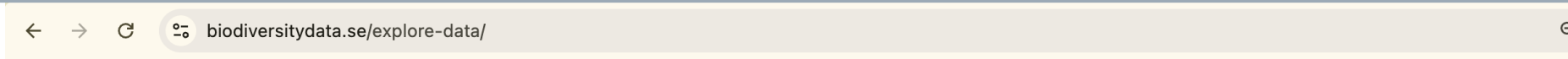
Publishers



© OpenStreetMap contributors, © OpenM

→ [Explore on gbif.org](https://gbif.org)

We provide tools and services



Swedish Biodiversity Data Infrastructure

HOME

EXPLORE DATA

PUBLISH DATA

EVENTS

SUPPORT

ABOUT

Q

Explore data

Tools and data

Programming and
API:s

Acknowledging SBDI

Tools and data

Here we list and describe all SBDI tools and portals that can be used to explore, download, and analyze biodiversity data. Our core tools and data access functionality is based on Living Atlases technology, but we also provide several specialized tools which are also linked to below.

You can filter the list by selecting one of the tags:

Search biodiversity datasets

Spatial analysis

Molecular/genetic data

Systematic inventories

Marine data

Biodiversity data

Archaeological data

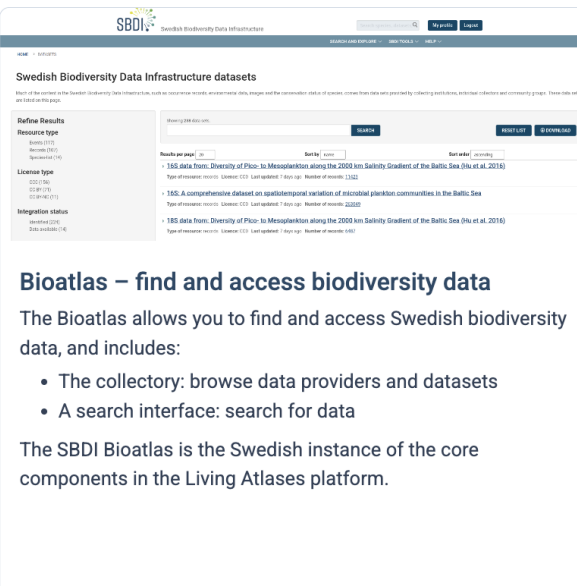
Data flows

Visualize/explore

Project management

Programming

SBDI Bioatlas

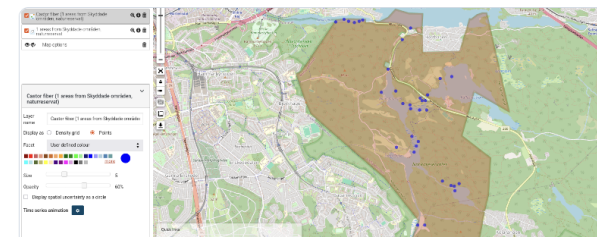


Bioatlas – find and access biodiversity data

The Bioatlas allows you to find and access Swedish biodiversity data, and includes:

- The collectory: browse data providers and datasets
- A search interface: search for data

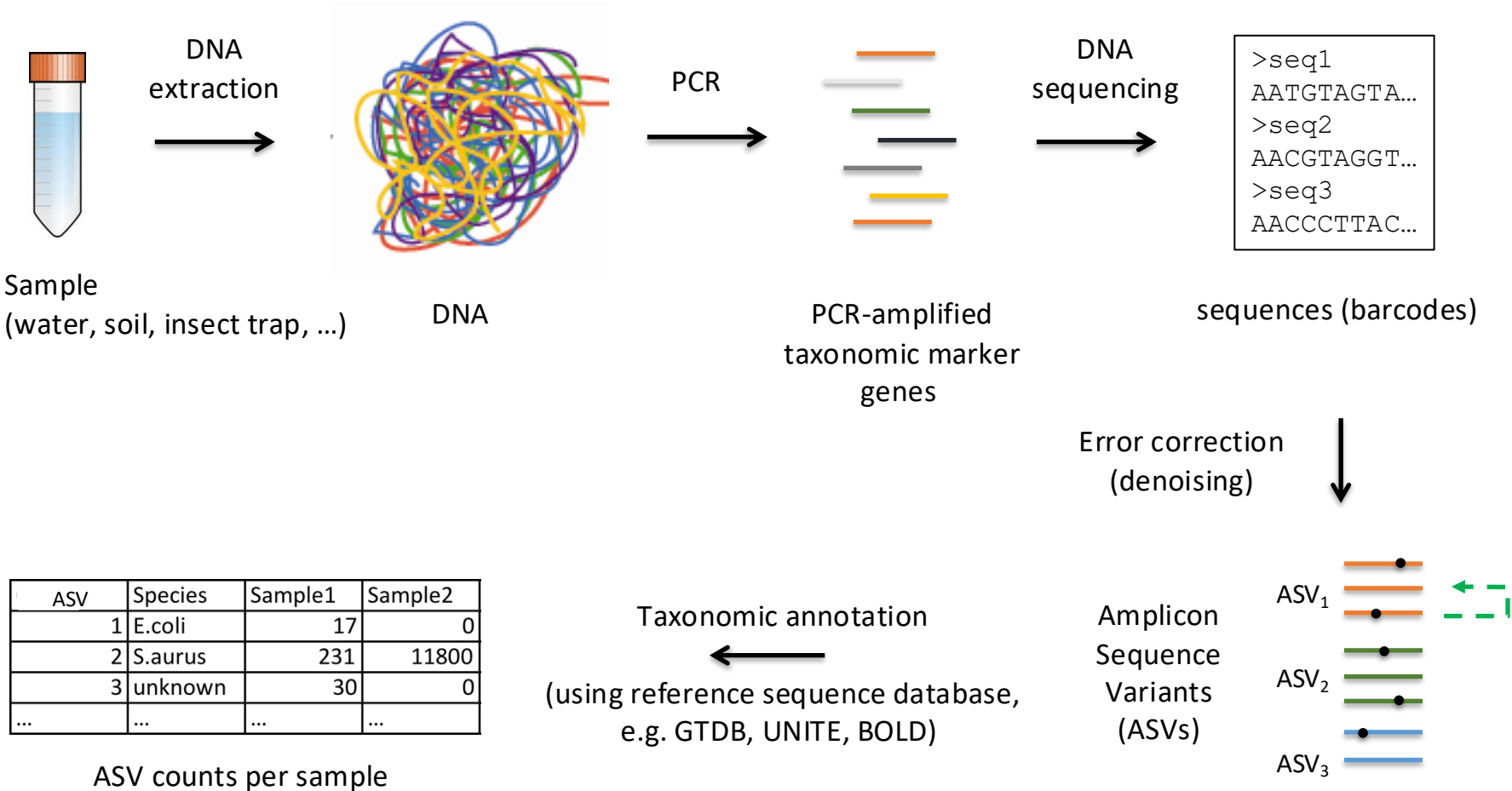
The SBDI Bioatlas is the Swedish instance of the core components in the Living Atlases platform.



Spatial Portal – visualize and analyze biodiversity data

The Spatial Portal is a tool that allows users to visualize and analyze Swedish biodiversity data in a spatial context – to understand and analyze biodiversity and its relationship with the environment.

- Map and analyze species occurrences alongside environmental and contextual data: map species distributions, explore relationships between species and environment, and conduct spatial analyses

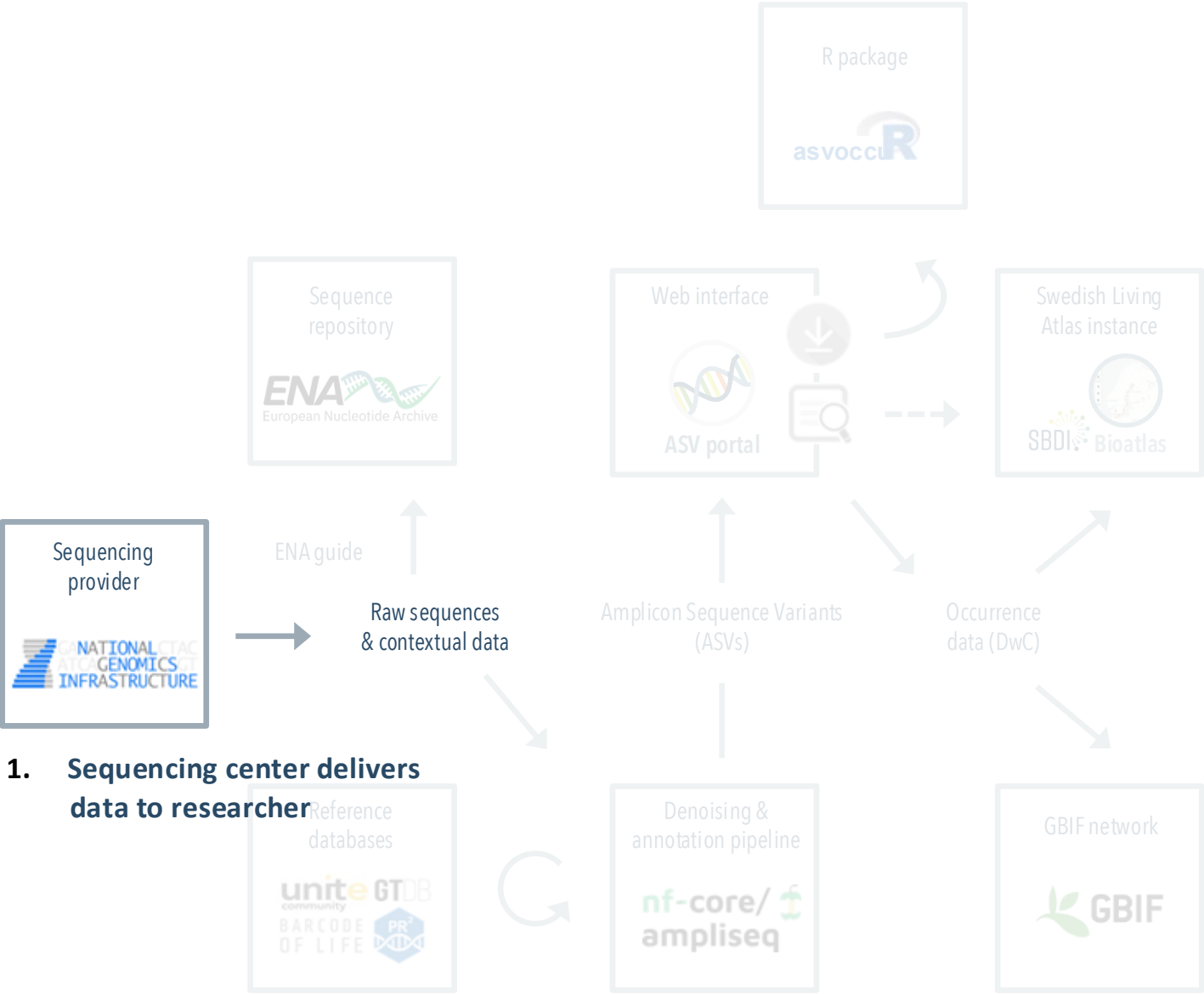




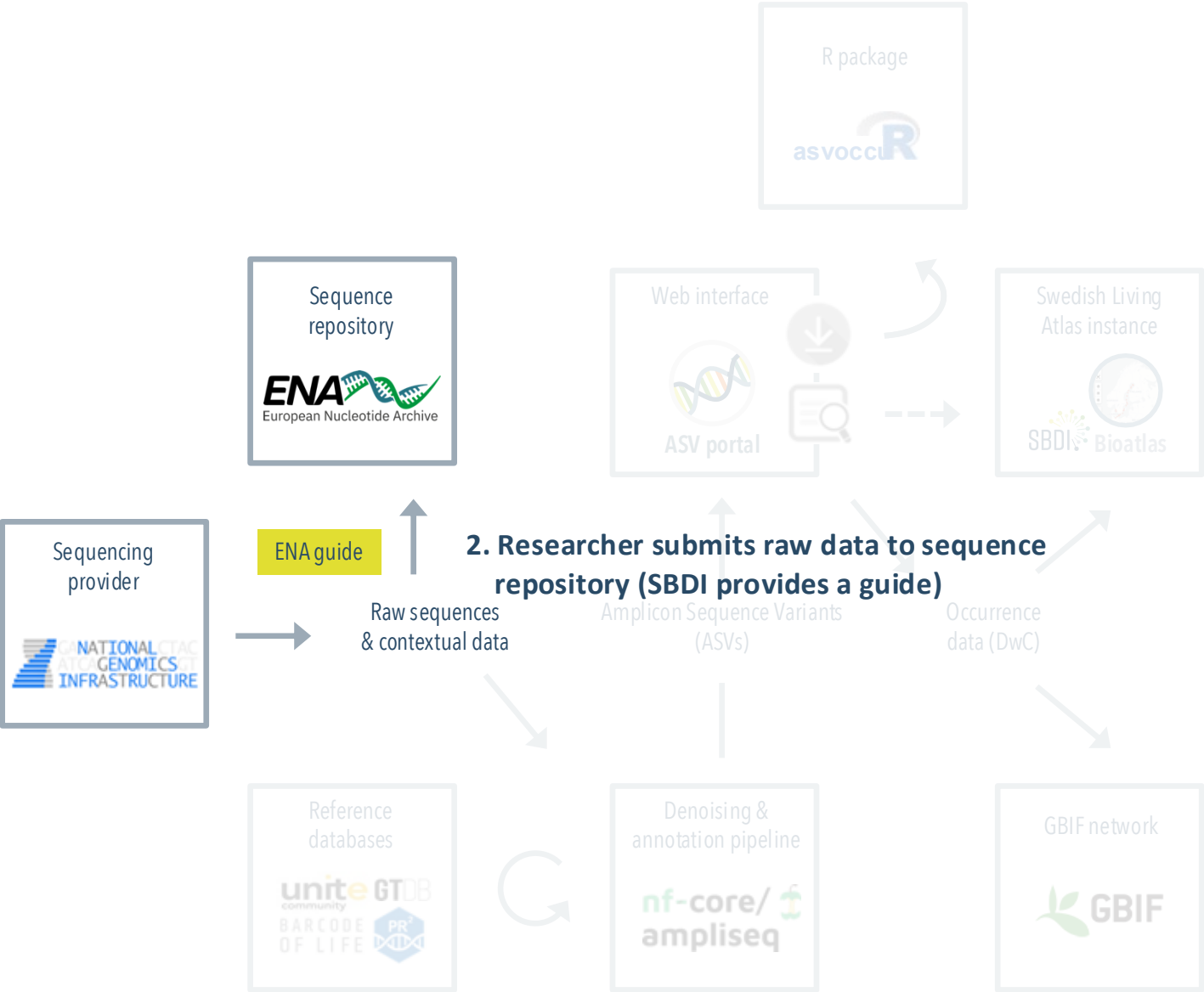
eDNA in SBDI

- tools and data flow...

eDNA in SBDI: tools & data flow



eDNA in SBDI: tools & data flow



🏠 Swedish Biodiversity Data Infrastructure: Molecular Data

Search docs

📁 Guide to ENA submission (webin)

📁 Preparation for submission

📁 Interactive submission

Step 1: Log in to submission portal

Step 2: Register study

📁 Step 3: Register samples

📁 Step 4: Prepare and upload read files

📁 Step 5: Submit sequence reads

Step 6: Submit to production service

Post-submission editing

Taxonomic annotation of ASVs

Step 3: Register samples

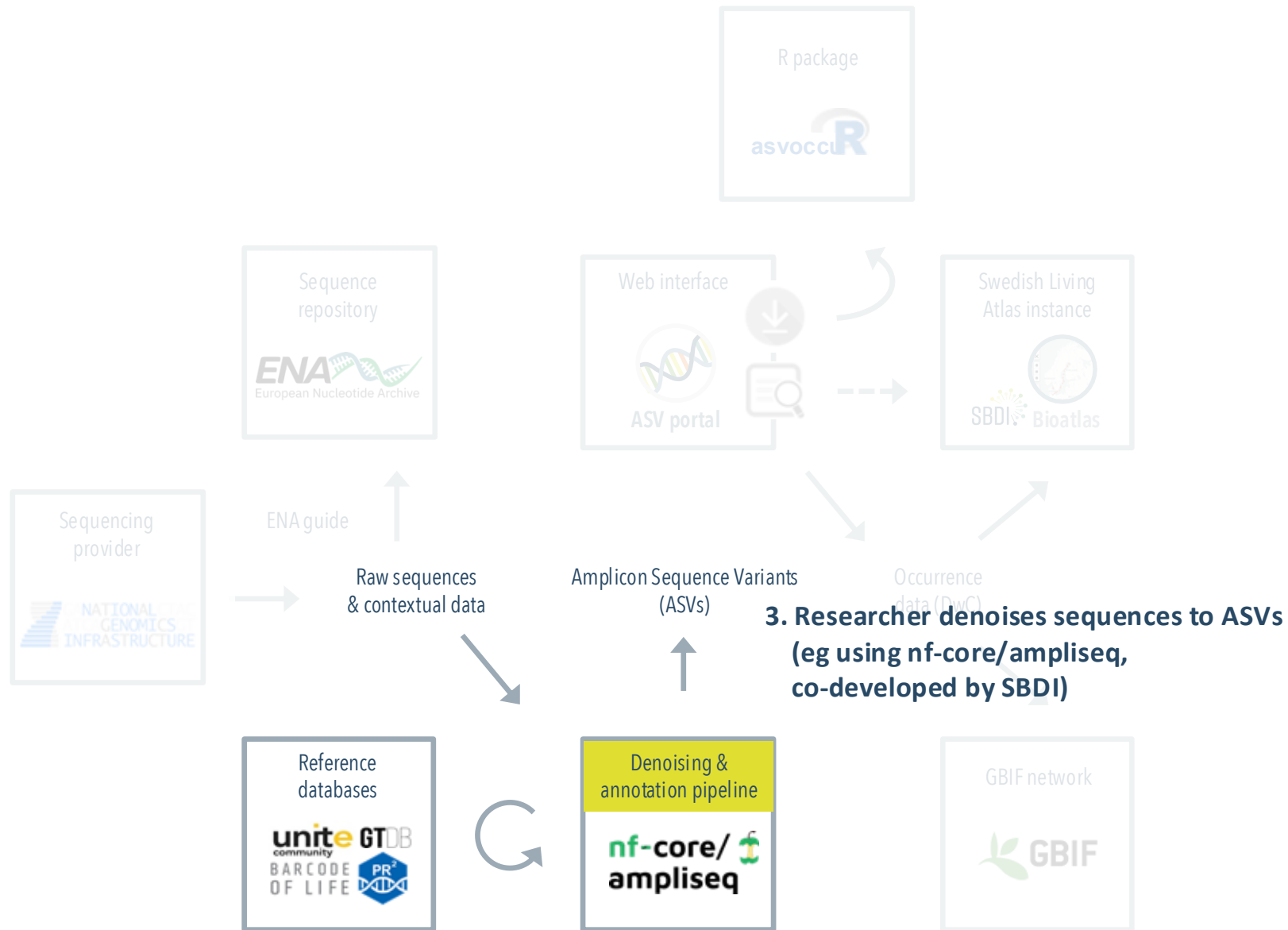
Samples are the source material from which your sequences derive, and the searchability and usability of your submitted data will depend on how well you document these samples. Go to *Samples* | *Register Samples* and click *Download spreadsheet to register samples* to start the process.

Step 3a: Select sample checklist

The [ENA sample checklists](#) are partly overlapping sets of attributes (or data fields) that can be used to describe samples, and by selecting one of these you enable your sample metadata to be validated for correctness during submission. For environmental and organismal (host-associated) samples, alike, we recommend using one of the *Environmental Checklists* and, among these, to select the alternative from the *Genomic Standards Consortium (GSC) MlxS checklists* that provides the most specific match to your sampled environment, for example:

Sampled environment	Recommended checklist
Air or general, above-ground, terrestrial	GSC MixS air
Epi- or endophytic (e.g. leaf, root)	GSC MlxS plant associated
Epi- or endozoic (e.g. spider gut, animal skin)	GSC MlxS host associated
Fresh- or seawater	GSC MixS water
Human gut / oral / skin / vaginal	GSC MlxS human gut / oral / skin / vaginal
Human non- gut / oral / skin / vaginal	GSC MlxS human associated
Sediment	GSC MixS sediment
Soil	GSC MixS soil

eDNA in SBDI: tools & data flow



nf-core/ampliseq

Amplicon sequencing analysis workflow using DADA2 and QIIME2

16s 18s amplicon-sequencing edna illumina iontorrent its metabarcoding metagenomics metataxonomics microbiome pacbio qiime2 rrna taxonomic-classification taxonomic-profiling

Launch version 2.18.0

<https://github.com/nf-core/ampliseq>

Introduction

Usage

Parameters

Output

Results

Introduction

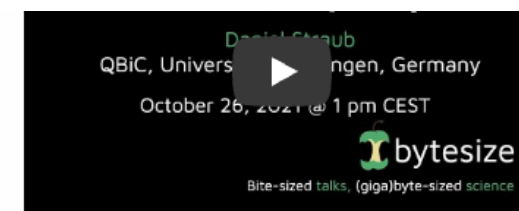
nfcore/ampliseq is a bioinformatics analysis pipeline used for amplicon sequencing, supporting denoising of any amplicon and supports a variety of taxonomic databases for taxonomic assignment including 16S, ITS, CO1 and 18S. Phylogenetic placement is also possible. Multiple region analysis such as 5R is implemented. Supported is paired-end Illumina or single-end Illumina, PacBio and IonTorrent data. Default is the analysis of 16S rRNA gene amplicons sequenced paired-end with Illumina.

A video about relevance, usage and output of the pipeline (version 2.1.0; 26th Oct. 2021) can also be found in [YouTube](#) and [billibili](#), the slides are deposited at [figshare](#).

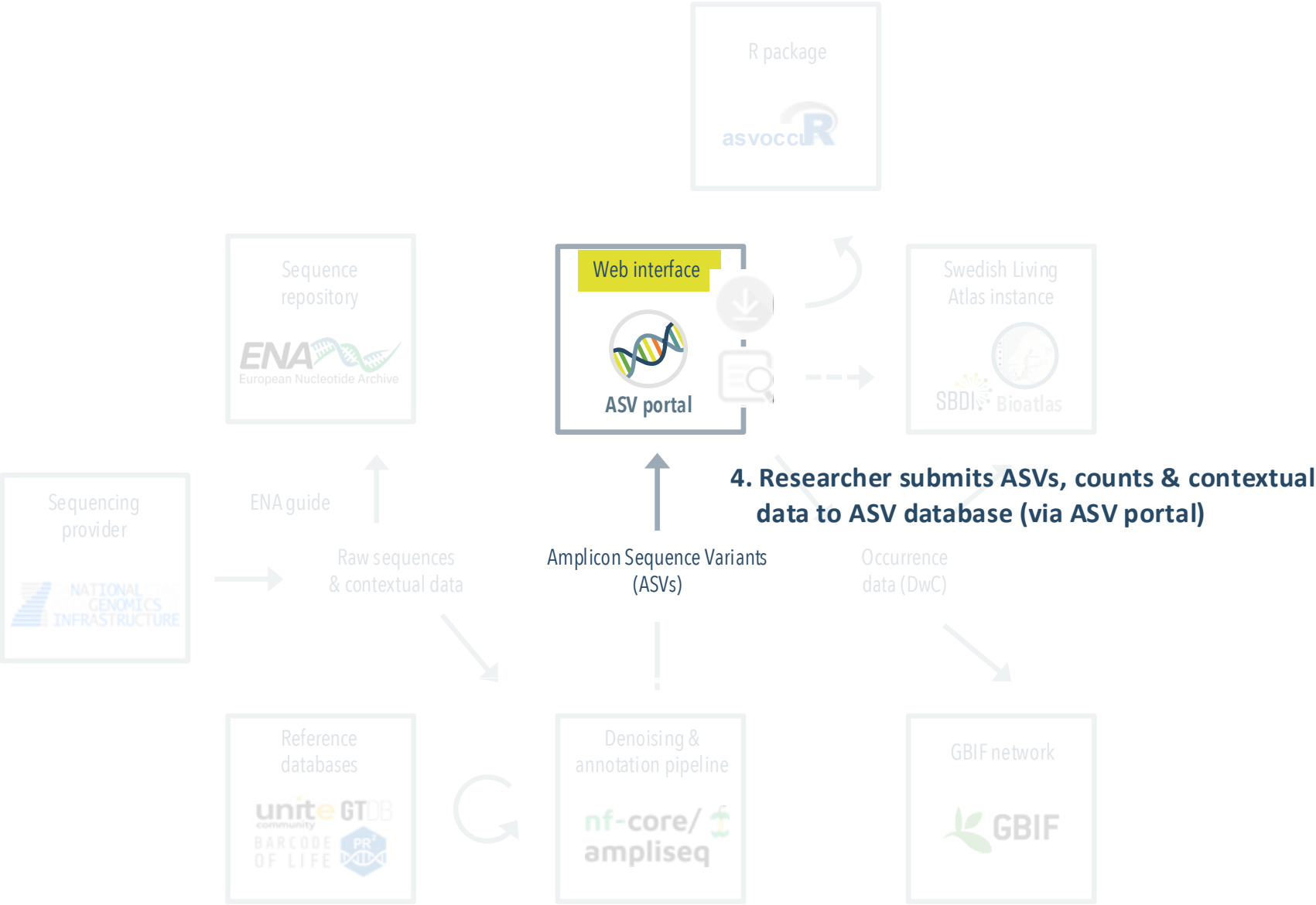


Daniel Lundin (LnU)

See ampliseq workshop
13-14:30 today



eDNA in SBDI: tools & data flow



Swedish ASV portal

Explore and share DNA sequences and species occurrences derived from metabarcoding (eDNA) studies



Search for ASVs and Bioatlas records using Basic Local Alignment Search Tool (BLAST)

BLAST

Search for ASVs and Bioatlas records using filters on sequencing details and taxonomy

FILTER



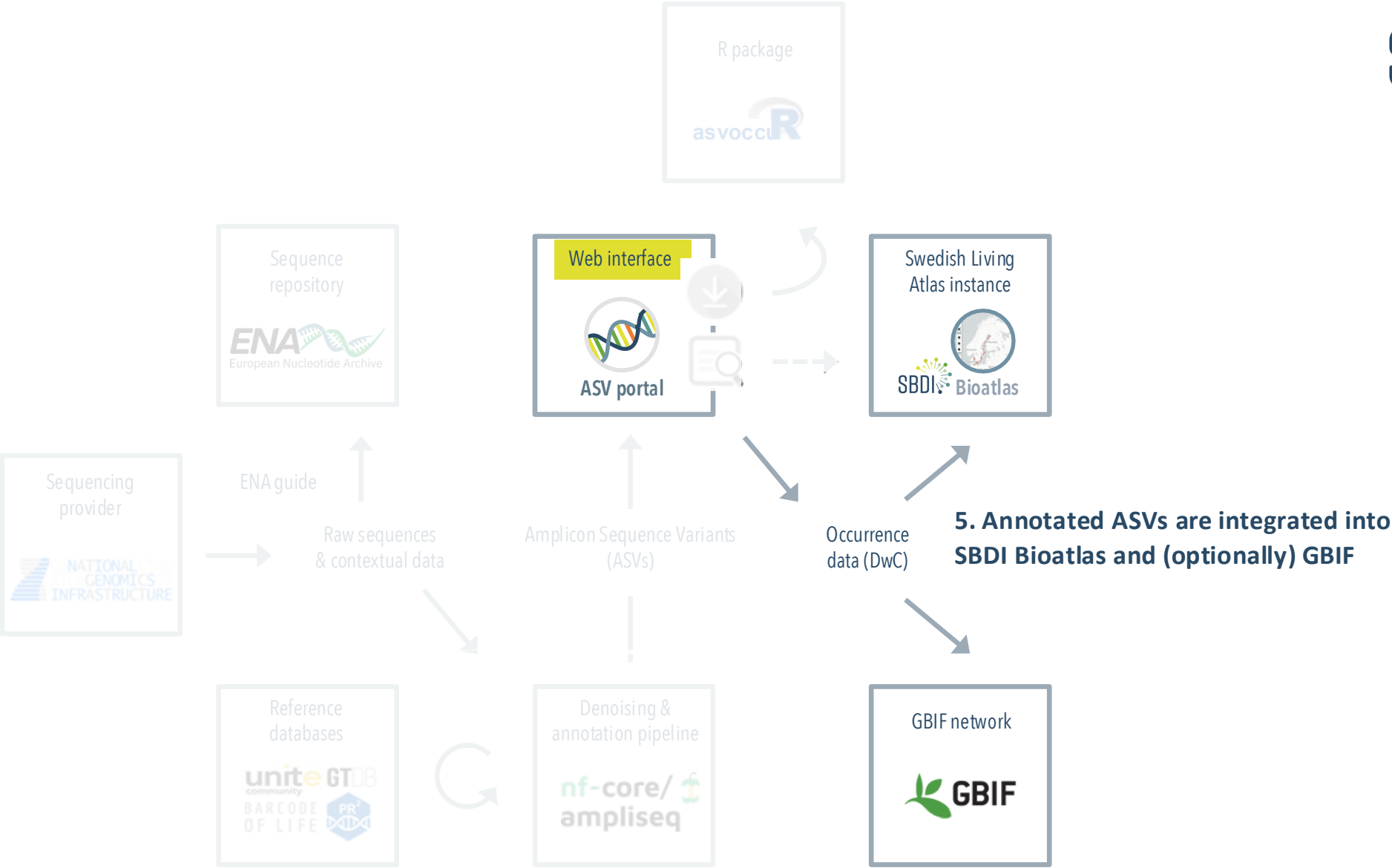
Submit your metabarcoding dataset to the ASV database and SBDI Bioatlas

SUBMIT

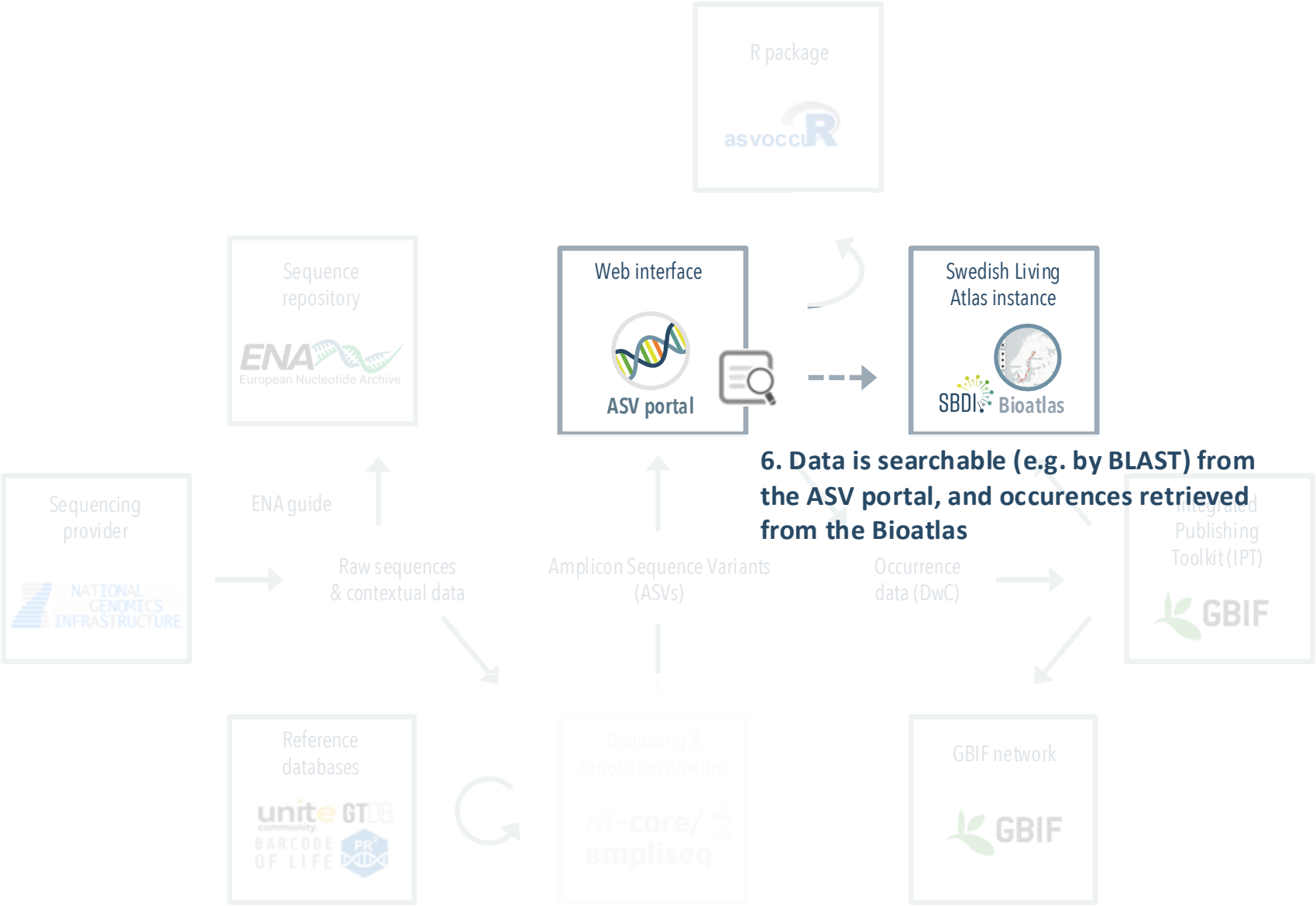
Download ASV occurrence datasets, in a condensed Darwin Core-like format

DOWNLOAD

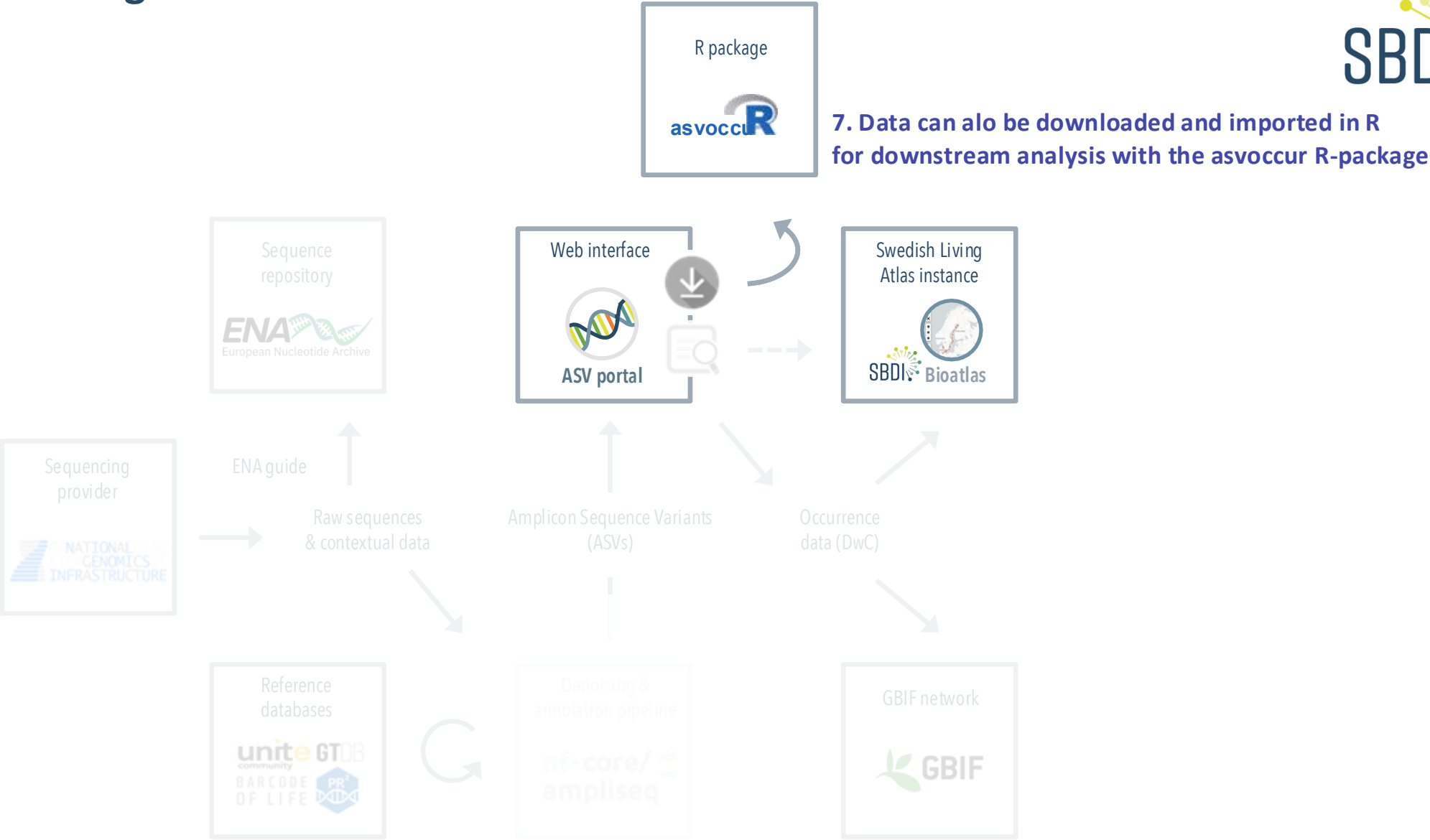
eDNA in SBDI: tools & data flow



eDNA in SBDI: tools & data flow



Metabarcoding data in SBDI: Dataflow



asvoccu

R tools for ASV occurrence data in [SBDI](#).

See [Releases](#) for the changelog.

Overview

The **asvoccu** R package, currently under development, provides tools for unpacking and processing ASV occurrence data and metadata downloaded from [the Swedish ASV portal](#). It enables users to convert condensed DwC archives into ASV table format for easier downstream analysis in R, by using functions that load, merge, and aggregate ASV counts across taxonomic ranks.

Tutorial

See our [tutorial](#) for an example workflow using **asvoccu** with ASV portal data in R.

Install



```
install.packages('remotes')
remotes::install_github("biodiversitydata-se/asvoccu")
# or:
# remotes::install_github("biodiversitydata-se/asvoccu@develop")
library(asvoccu)
```



Why publish eDNA data in SBDI?

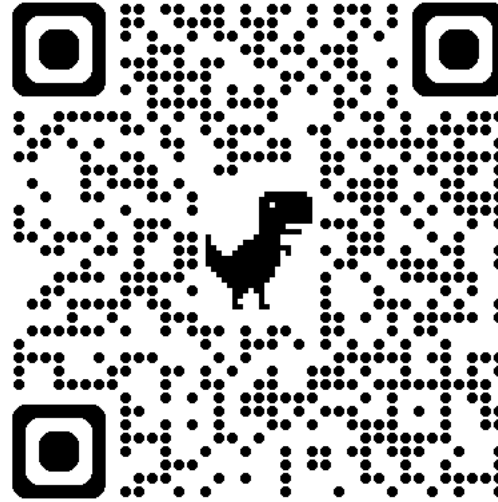


We help you make your data:

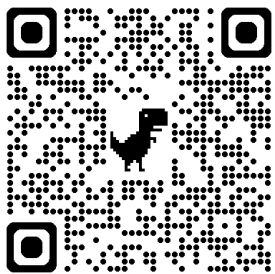
- FAIR & reusable – enable reuse by others
- Comparable – combine with other Bioatlas data
- Citable* – get a DOI for your dataset
- Up to date – benefit from regular ASV reannotation
- Globally accessible* – contribute to biodiversity knowledge through GBIF

*If you opt for GBIF export

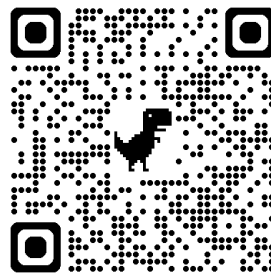
Today's tutorial
(& PP slides)



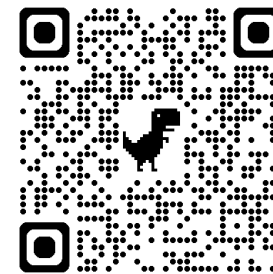
ENA submission guide



Pipeline: nf-core/ampliseq



Web interface: ASV portal



R-package: asvoccu

